



SALAS O'BRIEN
3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Falcon Landing; D.R. Horton Atlanta West
Location: Gainesville, GA
Description: Belford - D2/D3, Slab, ACH50

Date: September 22, 2024
By: TJH
House Faces: Worst East

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,248 ft ²
Volume	20,049 ft ³
Percentage Glass	11 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	8727
SEER	15
ACH50	3.0
Mechanical Ventilation	YES
House Outside Air CFM	75

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x6	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.5	Exterior Door		R-2	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling	2x4	R-38 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.35	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		26864 Btu/hr		31148 Btu/hr	
Minimum Nominal Capacity		2.98 Tons		9.13 kW	
Load Density		753 SF/Ton		4.1 W/SF	
Floor Area Served		2248.0 SF		2248.0 SF	
Mechanical Ventilation		75 CFM		75 CFM	
System Size		3.0 Tons		15.0 kW	
Leaving Air Temperature		60.1 deg F		103.9 deg F	
Installed Capacity		749 SF/Ton		6.7 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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Unit 1

Size

3.00

Tons

Airflow

1200

CFM

Heat kW

15

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer & Closet	32	56	44	
Hall	9	11	10	
Powder	3	5	4	
Pantry	2	3	3	
Bed 1 WIC	10	21	15	
Bath 1	29	52	41	
Bed 1	126	140	133	
Great Rm	251	235	243	
Laundry	41	19	30	
Garage Entry Hall	24	38	31	
Kitchen	154	54	104	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	67	123	95	
Landing / Hall	51	55	53	
Bed 4 WIC	6	9	8	
Bed 4	69	67	68	
Upstairs Living	113	48	81	
Bed 3	74	83	79	
Bath 2	25	39	32	
Bed 2 WIC	25	39	32	
Bed 2	90	101	95	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	

DX Cooling kWh	1897
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By: TJH

RC Heating kWh	2850
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Level(s): First Floor

Total kWh	4747
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SEER 15

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DX Cooling kWh	1446
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RC Heating kWh	2533
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Total kWh	3979
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SEER 15

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